

Git Helper — User Guide

Git Helper is a friendly Git client for **Azure DevOps**. It explains your repo's state in plain language, shows the next safe step, and covers the whole loop: **connect** → **clone** → **change** → **commit** → **push** → **pull request** — without the Git CLI.

Get the app: download the installer for Windows or macOS from the [latest release](#), or run it from source (`npm run dev` → `http://localhost:5173`). First launch on Windows: **More info** → **Run anyway**; on macOS: **right-click** → **Open**.

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1. Connect to Azure DevOps

When the app opens without a connection you land on the **Connect** screen.

GIT, IN PLAIN LANGUAGE

Know exactly where your repo stands — and what's safe to do next.

A calmer way to work with Azure DevOps Repos. No CLI flags, no guessing whether it's safe to pull. Every screen tells you where you are, what the state is, and your next safe step.

- Green means up to date and safe to act.
- Amber means something needs your attention first.
- Red means blocked or in conflict — handle with care.

Connect to Azure DevOps

Your token is sent to this app's local server and kept in your session — never stored in the browser.

Organization

From dev.azure.com/fabrikam

Project

Personal access token

[How do I create a token?](#)

Connect

Prototype mode · read-only access is enough to explore

OR

Work with a local repository

No token needed. Open a Git folder on this machine to visualise history, stage and commit, and resolve merge conflicts — all in plain language.

Repository folder

Choose a folder...

Pick a folder on this machine that contains a Git repository.

Fill in three things:

Field	What to enter
Organization	The name after <code>dev.azure.com/</code> in your browser — e.g. for <code>dev.azure.com/fabrikam</code> enter <code>fabrikam</code>
Project	The project name exactly as it appears in Azure DevOps
Personal access token	A PAT — see below

Creating a PAT (one time): in Azure DevOps click your avatar (top right) → **User settings** → **Personal access tokens** → **New Token**. Give it a name, pick your organization, and under **Scopes** choose **Code** → **Read & write** (read-only is enough for browsing, but cloning private repos, pushing, and completing pull requests need write). Copy the token and paste it into the app.

Your token is kept in an **encrypted session cookie on your machine** — it is never stored in the browser or sent anywhere except Azure DevOps.

Press **Connect**. The sidebar now shows the **Azure DevOps** section (Dashboard, Repositories, Branches, Compare branches, Pull requests).

Prefer to work without Azure? Use **Work with a local repository** below the form — pick any Git folder on your machine and skip to [Status](#).

Connecting later, and disconnecting

You don't have to connect right away — the two modes are independent:

- **+ Connect Azure DevOps** (bottom of the sidebar) — if you started with a local repository only, click this any time to open the Connect screen and add the Azure side. Both sections then live together in the sidebar, and cloning from [Repositories](#) becomes available.
- **Disconnect Azure** (bottom of the sidebar) — signs you out and forgets the token for this session. Your local repositories are untouched.
- **Close local repo** — deselects the current local repository (it stays in [Your repositories](#) for one-click reopening).

The connection lasts for your session (about 8 hours) — after that, just connect again with the same PAT until the token itself expires.

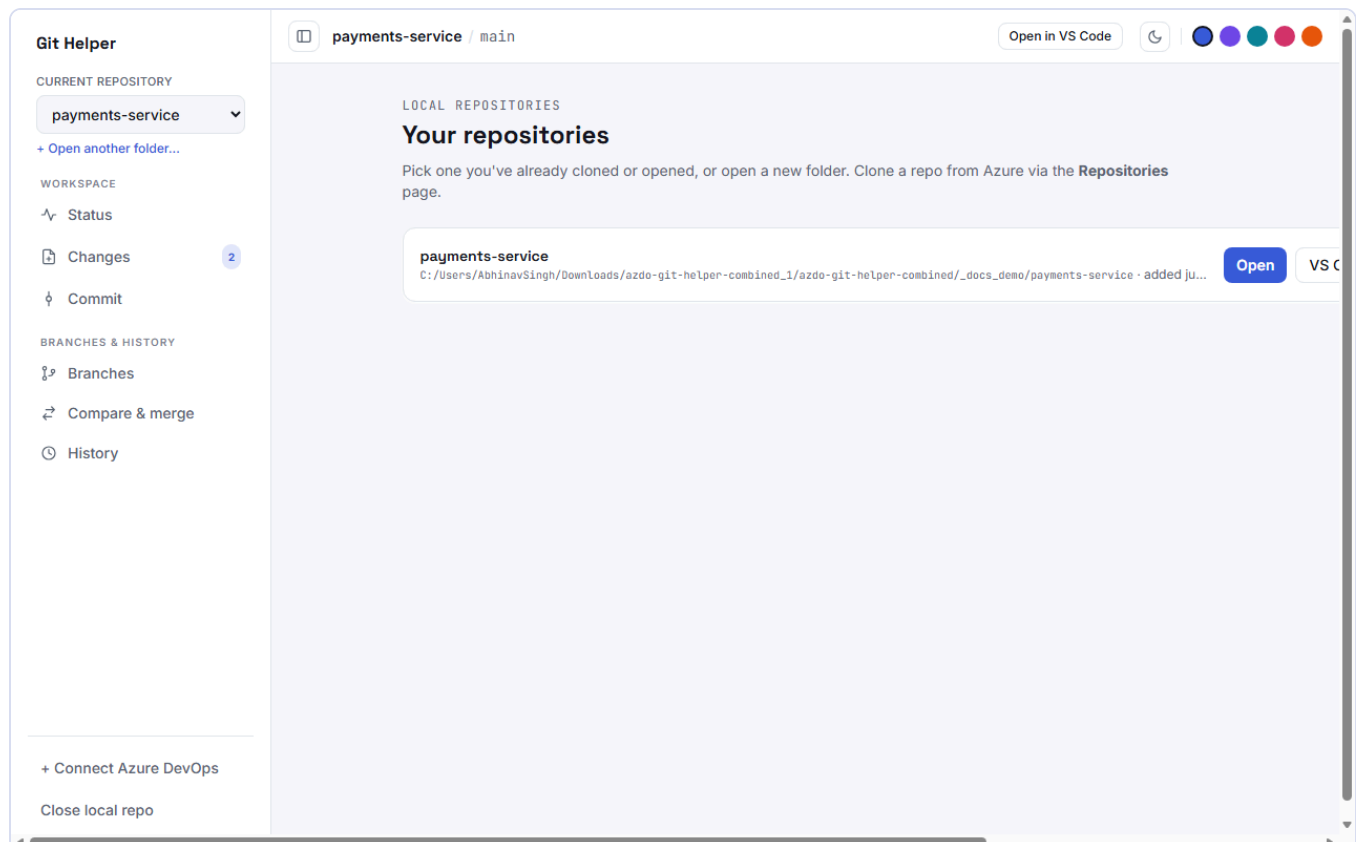
2. Clone a repository

1. In the sidebar under **Azure DevOps**, open **Repositories**.
2. Every repo in your project is listed. Click **Clone to local** on the one you want.
3. A dialog asks **where** to put it — click **Choose...** to browse your disk (you can jump between drives, go **Up**, or **Home**), then **Clone here**.
4. A **progress bar appears at the top** of the app while Git downloads (`Receiving objects... 45%`).
5. When it finishes, the repo **opens automatically** in the local workspace — you land on [Status](#), already wired to Azure as its `origin`.

You only ever clone a repo **once** — after that it stays in [Your repositories](#).

3. Your repositories

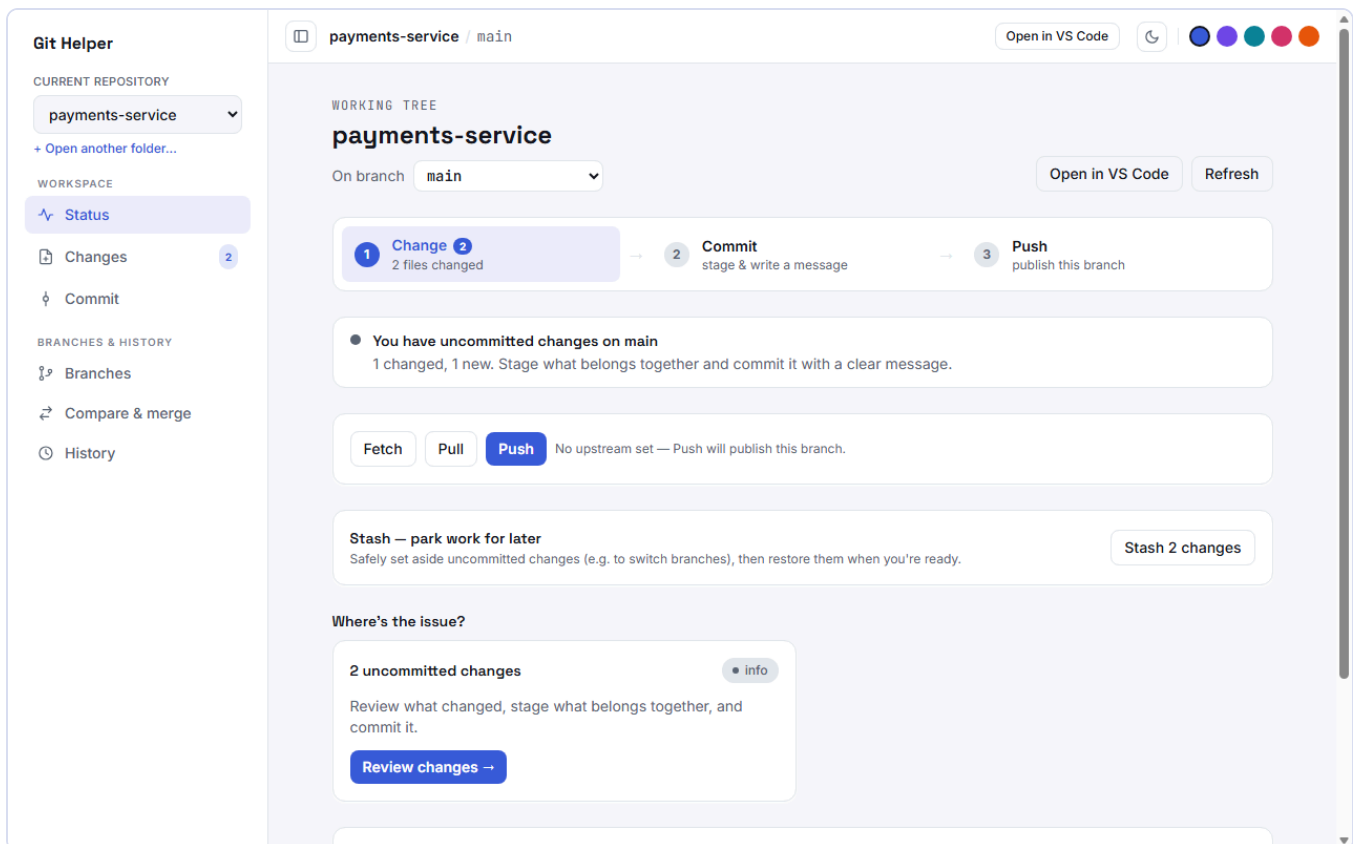
Everything you've cloned or opened is remembered. Switch repos any time with the **Current repository** dropdown at the top of the sidebar, or manage the full list via **+ Open another folder....**



- **Open** — makes it the current repo.
- **VS Code** — opens that repo's folder in Visual Studio Code.
- **X** — removes it from the list (the files on disk are *not* deleted).
- **Open a folder** — add any existing Git repo from your machine using the built-in folder browser.

4. Status — your home screen

Whenever you open a repo you land here. It answers: *where am I, what's the state, and what should I do next?*



Top to bottom:

- **On branch** — shows the current branch. The dropdown **switches branches** right here (remote branches are listed too — picking one checks it out locally).
- **The 1-2-3 strip** — your workflow at a glance: **Change** → **Commit** → **Push**, with live counts. The highlighted step is your next move; completed steps show ✓. Each step is clickable.
- **The banner** — plain-language summary (green = safe, amber = needs attention, red = blocked).
- **Fetch / Pull / Push** — talk to Azure DevOps. The counter shows how many commits you're ahead ↑ / behind ↓.
- **Stash** — park uncommitted work safely ([details](#)).
- **Where's the issue?** — cards that point you at whatever needs doing, with a button straight to the fix.

Top-right: **Open in VS Code** and **Refresh**. The navbar also shows the current repo/branch and the [appearance controls](#).

5. Changes — review what you did

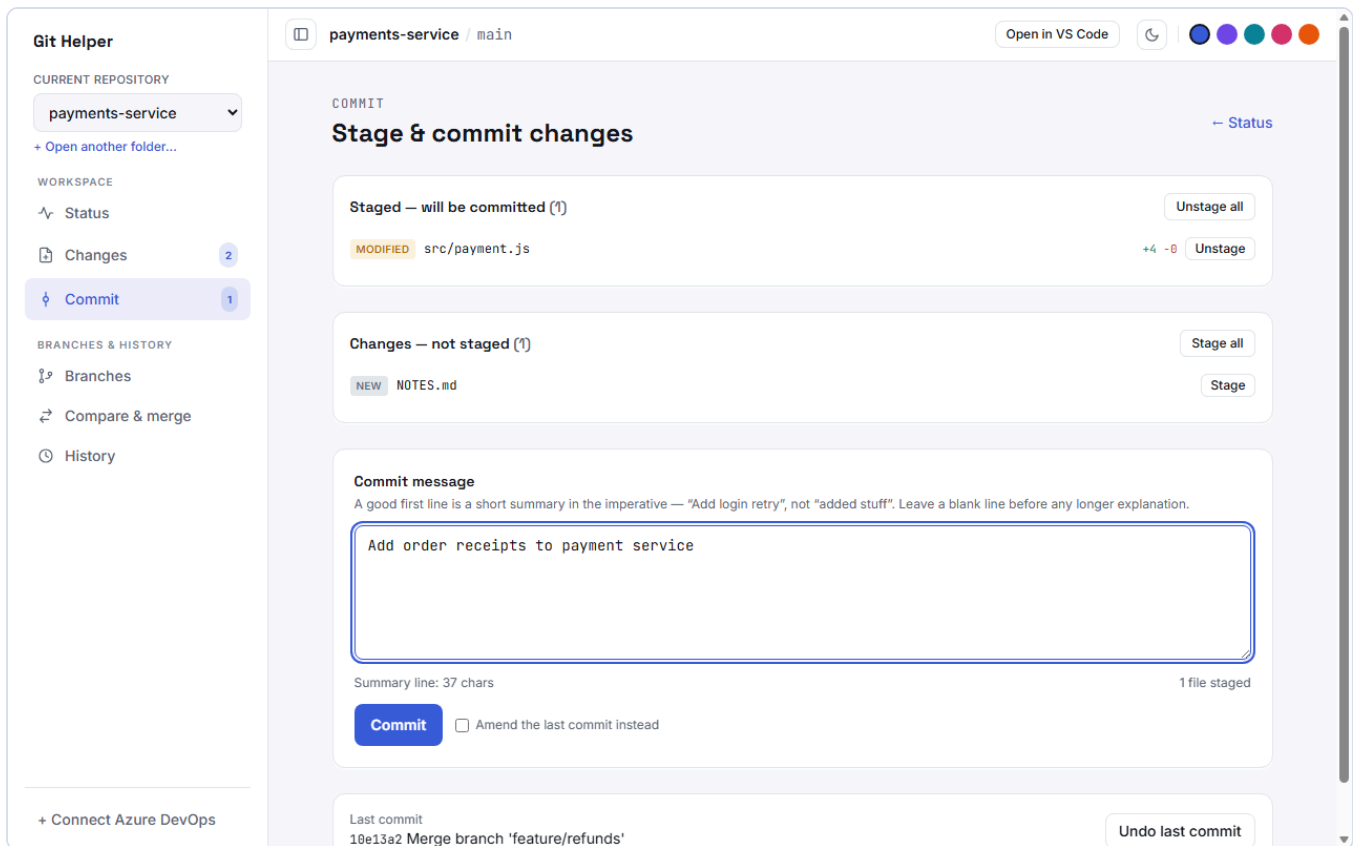
Sidebar → **Changes** (the badge shows how many files changed).

The screenshot shows the Git Helper application interface. On the left is a sidebar with sections: CURRENT REPOSITORY (payments-service), WORKSPACE (Status, Changes, Commit), BRANCHES & HISTORY (Branches, Compare & merge, History), and a bottom section with '+ Connect Azure DevOps' and 'Close local repo'. The main area is titled 'payments-service / main' and shows 'Working-tree changes' with '2 files changed · +4 -0'. A progress bar at the top indicates three steps: 1. Change (2 files changed), 2. Commit (stage & write a message), and 3. Push (publish this branch). Below this, a file 'src/payment.js' is shown as 'MODIFIED' with a diff view. The diff shows a function 'charge' being modified and a new function 'receipt' being added. A 'Discard' button is next to the file. At the bottom, a 'NEW' file 'NOTES.md' is shown as 'UNTRACKED' with a 'Discard' button. A tip at the bottom says: 'Tip: commit these on a branch, then push to publish to origin.'

- Every changed file is listed with **green +added / red -removed** counts and a label (modified / new / deleted / renamed).
- **Click a file** to expand its diff — green lines were added, red removed.
- **Discard** throws away that file's changes (it asks first — this cannot be undone).
- Conflicted files show a link straight to the [conflict resolver](#).
- **Stage & commit** → takes you to the Commit screen.

6. Commit — save your work

Sidebar → **Commit**.



1. **Stage** the files that belong together (per file, or **Stage all**). Staged files are what the commit will contain; click any file to double-check its diff. **Unstage** moves it back.
2. Write a message. Keep the first line a short summary (the counter warns past 72 characters) — e.g. *"Add order receipts to payment service"*.
3. Press **Commit**.

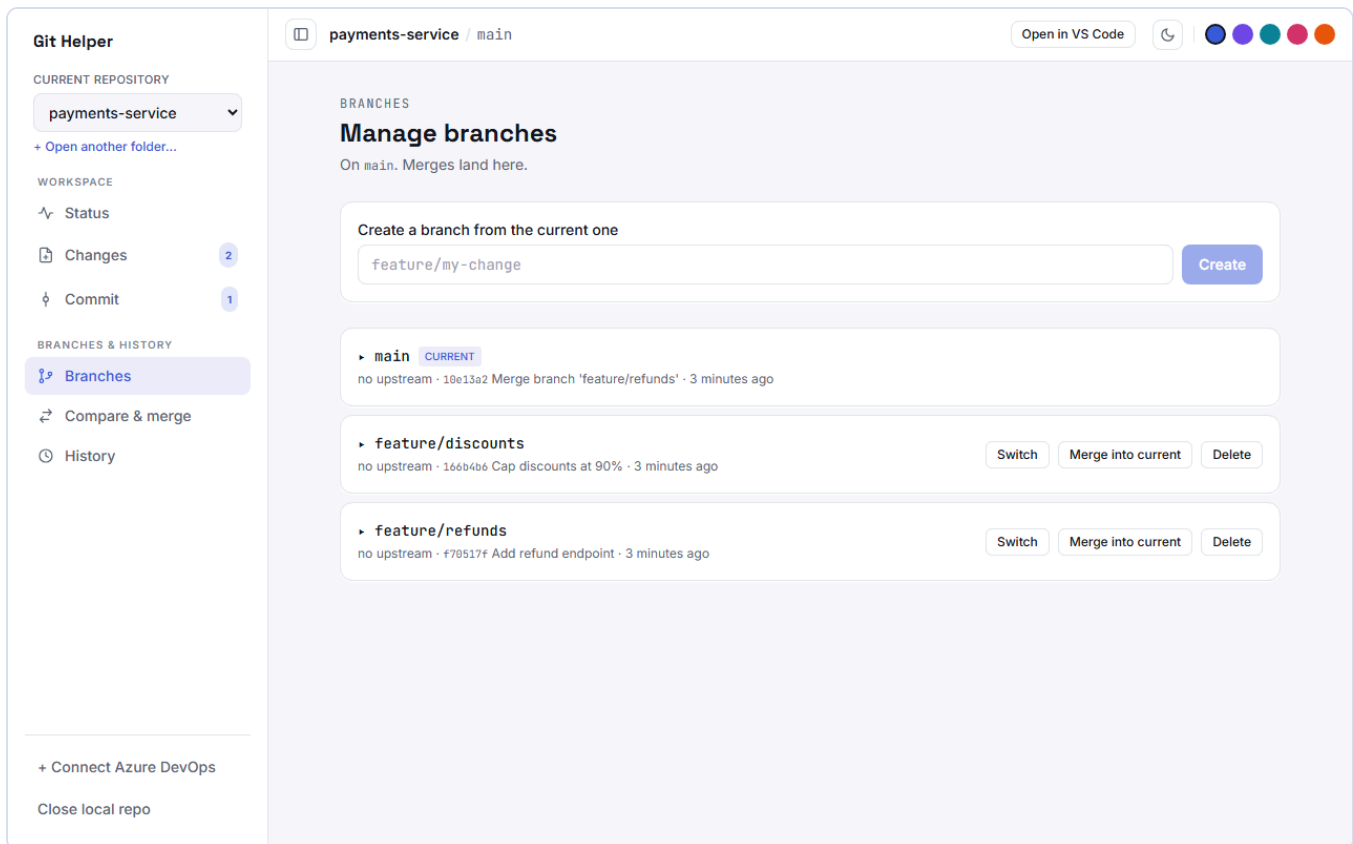
Two safety nets for the last commit (only while it hasn't been pushed):

- **Undo last commit** — takes the commit apart but **keeps all its changes staged**, so nothing is lost. Great for "oops, wrong files".
- **Amend the last commit instead** (checkbox) — replaces the last commit with the current staged files + message. Great for "forgot one file" or a typo in the message.

After committing, the strip moves to **Push** — [share it](#).

7. Branches

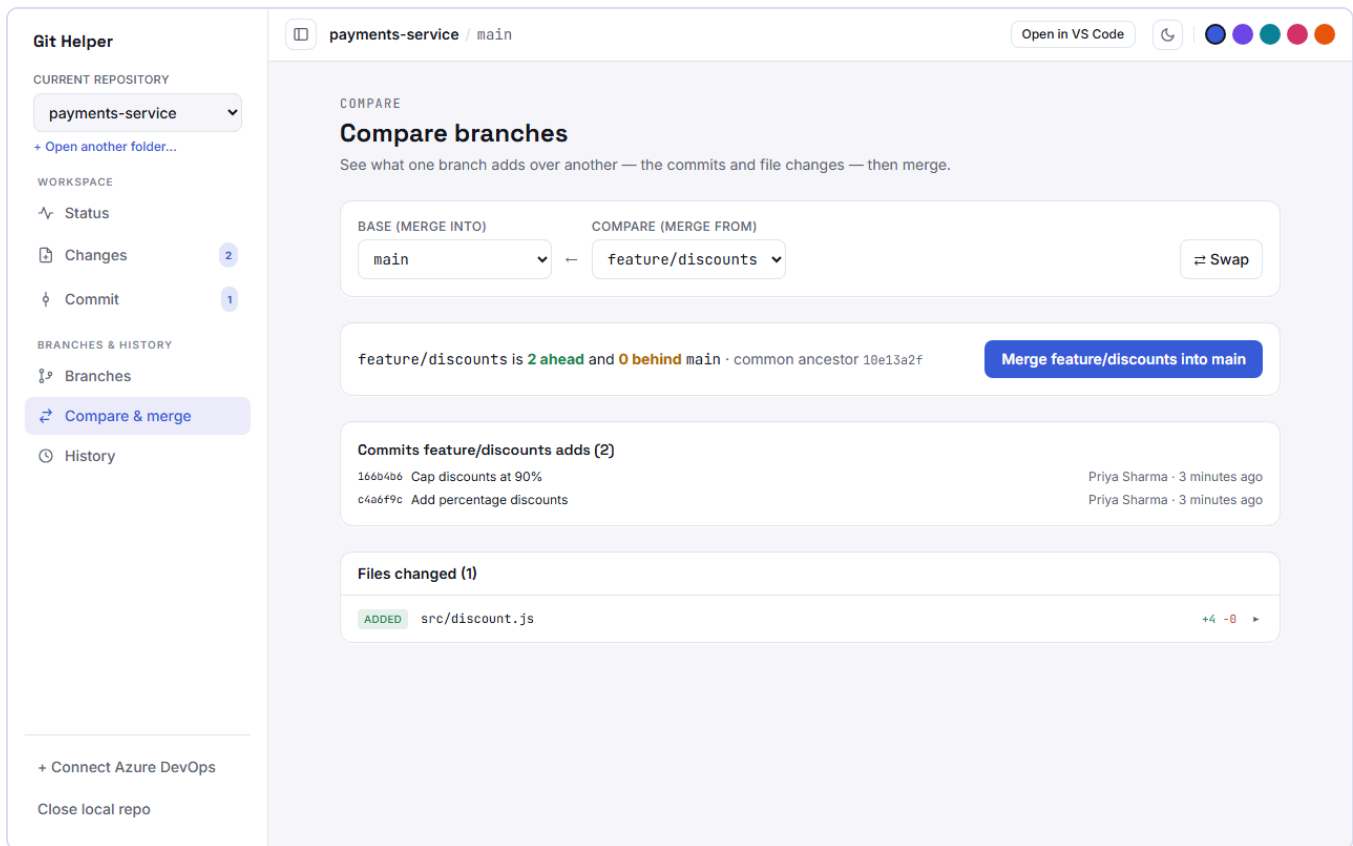
Sidebar → **Branches**.



- **Create a branch** — type a name (e.g. `feature/my-change`) and press **Create**; you're switched onto it immediately.
- Each branch row shows its last commit, and ahead ↑ / behind ↓ its upstream.
- **Switch** — check out that branch. Branches marked **remote** exist on Azure but not on your machine yet — **Check out** creates your local copy.
- **Merge into current** — merges that branch into the one you're on. If the same lines changed on both sides you're taken to the [conflict resolver](#).
- **Delete** — removes a local branch (Git refuses if it has unmerged work).
- Expand a row (▶) to see the commits that branch adds over the main branch.

8. Compare & merge

Sidebar → **Compare & merge**. See exactly what one branch would bring into another **before** merging — the best way to avoid surprise conflicts.



1. Pick a **Base** (merge *into*) and a **Compare** (merge *from*) — ⇌ **Swap** flips them.
2. You get: how many commits ahead/behind, the common ancestor, the exact **commit list**, and every **file changed** with expandable diffs.
3. One click merges: **Merge compare into base** (the app switches to the base branch for you if needed). Conflicts route into the resolver.

If the compare branch has *nothing new* and is just behind, the app says so in plain words and offers ⇌ **Flip it** — setting up the catch-up merge that branch actually needs.

9. History

Sidebar → **History**.

The screenshot shows the Git Helper application interface for the 'payments-service' repository. The sidebar on the left contains navigation options: 'CURRENT REPOSITORY' (payments-service), 'WORKSPACE' (Status, Changes, Commit), 'BRANCHES & HISTORY' (Branches, Compare & merge, History), and '+ Connect Azure DevOps'. The main area displays the 'History — payments-service' view, showing a commit history graph and a list of commits. A details panel on the right shows the merge of the 'feature/refunds' branch into 'main'.

Commit History:

- HEAD main Merge branch 'feature/refunds'** (10e13a2) - Priya Sharma - 3 minutes ago
- feature/refunds Add refund endpoint** (f70517f) - Priya Sharma - 3 minutes ago
- Add currency formatting helpers** (f397bf3) - Priya Sharma - 3 minutes ago
- Initial payment service** (137eac0) - Priya Sharma - 3 minutes ago

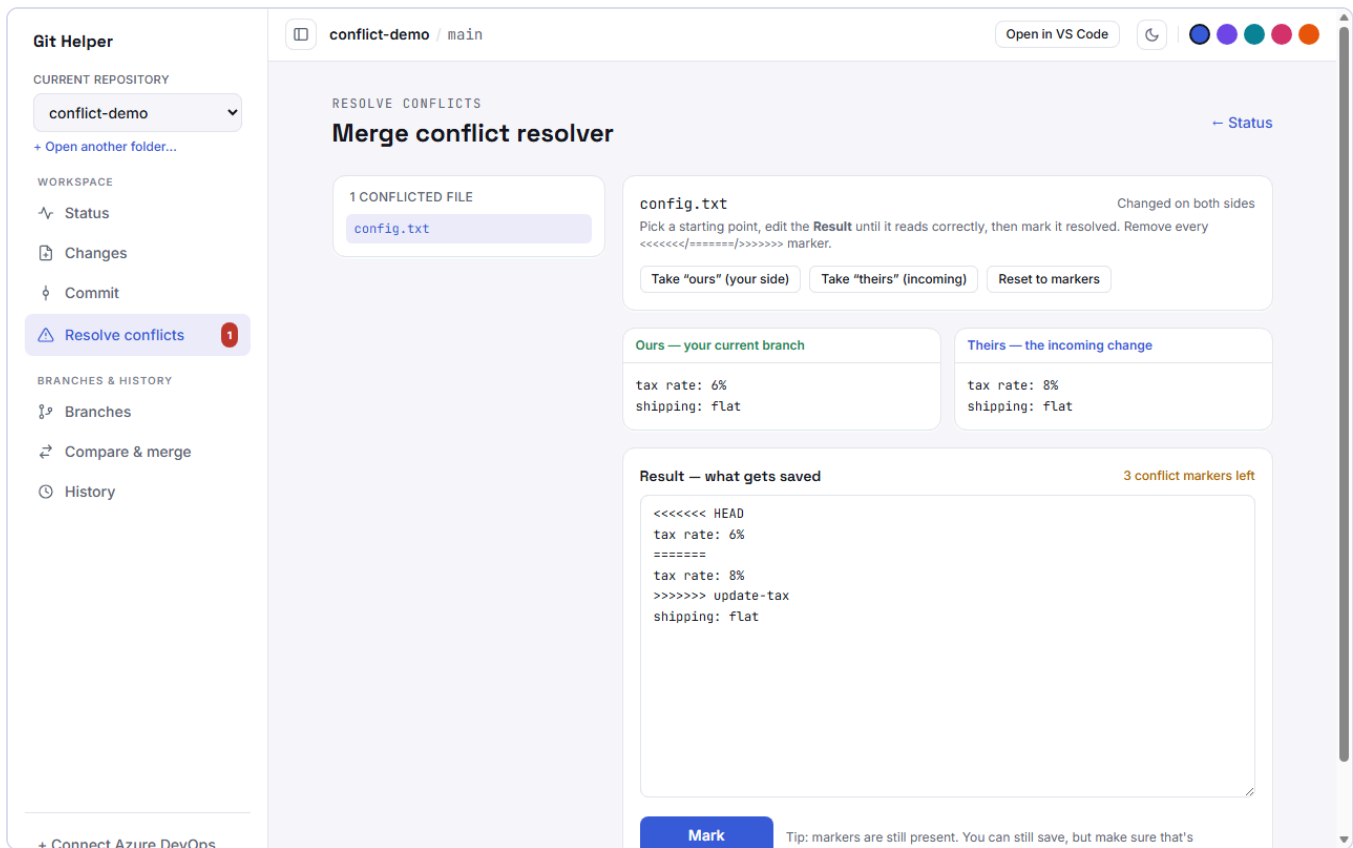
Details Panel:

commit 10e13a2f5746362f6acde6169bf13ce2cf0c1118
Merge branch 'feature/refunds'
 Priya Sharma <priya@contoso.com> · 3 minutes ago **MERGE**
 1 file changed +3 -0
 src/refund.js +3 -0

- **Graph | List** — the graph draws real branch/merge lanes; List is a clean linear feed.
- **Show** — the current branch (default), any specific branch, or **All branches**.
- **Simplify merges** — hides merged-in side branches and tells the branch's own story (great for busy shared branches).
- **Click any commit** → a details panel slides in from the right: full message, author, branch tags, and **every file it changed** — click a file for its diff. Click other commits to flip through them; **Esc** closes.

10. Resolve merge conflicts

When a merge/pull/stash hits a conflict, **Resolve conflicts** appears in the sidebar with a red count, and everything routes you here.



For each conflicted file you see three panes:

- **Ours** — how it looks on *your* branch.
- **Theirs** — the incoming change.
- **Result** — an editor holding what will be saved.

Pick a starting point (**Take "ours"**, **Take "theirs"**, or edit by hand to combine both), make the Result read correctly, and remove every <<<<<< / ===== / >>>>>> marker (the counter tracks them). Press **Mark resolved**, repeat for each file, then finish with a normal [commit](#).

11. Stash — park work for later

On [Status](#), when you have uncommitted changes:

- **Stash N changes** — sets *everything* aside (new files included) and leaves a clean tree. Perfect when you must switch branches mid-work.
- Each stash is listed with **Restore** (bring it back — conflicts route to the resolver) and **Delete**.

12. Push, Pull, Fetch

The toolbar on [Status](#):

Button	What it does
Fetch	Checks Azure for news (updates the ↑↓ counters; changes nothing locally)
Pull	Brings the upstream's commits into your branch (a conflict opens the resolver)
Push	Sends your commits to Azure. A brand-new branch is published automatically (sets its upstream)

Amber "behind" states are safe — pull first, then push.

Nothing runs by surprise. Each button first shows an "Are you sure?" dialog that spells out, for your exact situation, **what will happen** (e.g. "Your 2 local commits will be uploaded to origin/main") and **what it affects** (your files? your branch? the branch on Azure that teammates see?). The action only runs when you click **Yes** — Cancel or **Esc** changes nothing. If you try to push while you're also behind, the dialog warns you that Azure may reject it until you pull first.

13. Azure DevOps cloud features

These live in the sidebar's **Azure DevOps** section (after [connecting](#)) and work in the hosted web app too.

Dashboard

Pick a repo in the sidebar dropdown; the dashboard shows each branch's plain-language status vs the default branch — a divergence diagram and recent commits, with a traffic-light verdict.

Repositories

Every repo in the project, with default branch and links — plus **Clone to local** (see [cloning](#)).

Branches (cloud)

All branches on the server with ahead/behind counts and a verdict. **New PR** starts a pull request from any branch.

Compare branches (cloud)

Same idea as local [Compare & merge](#) but computed by Azure: pick base and target, see ahead/behind, common ancestor, and changed files. If the target is behind, the app warns you *before* you open a PR — that's where conflicts come from. **Open pull request** → carries your selection into the PR form.

Pull requests

Filter by **All / Created by me / Assigned to me** and by state. Open a PR to see its description, reviewers and their votes, and comment threads. On an active PR, **Complete this pull request** merges it on Azure: choose the strategy (merge / squash / rebase / semi-linear), optionally **delete the source branch**, and confirm. Policy blocks and conflicts are explained in plain language.

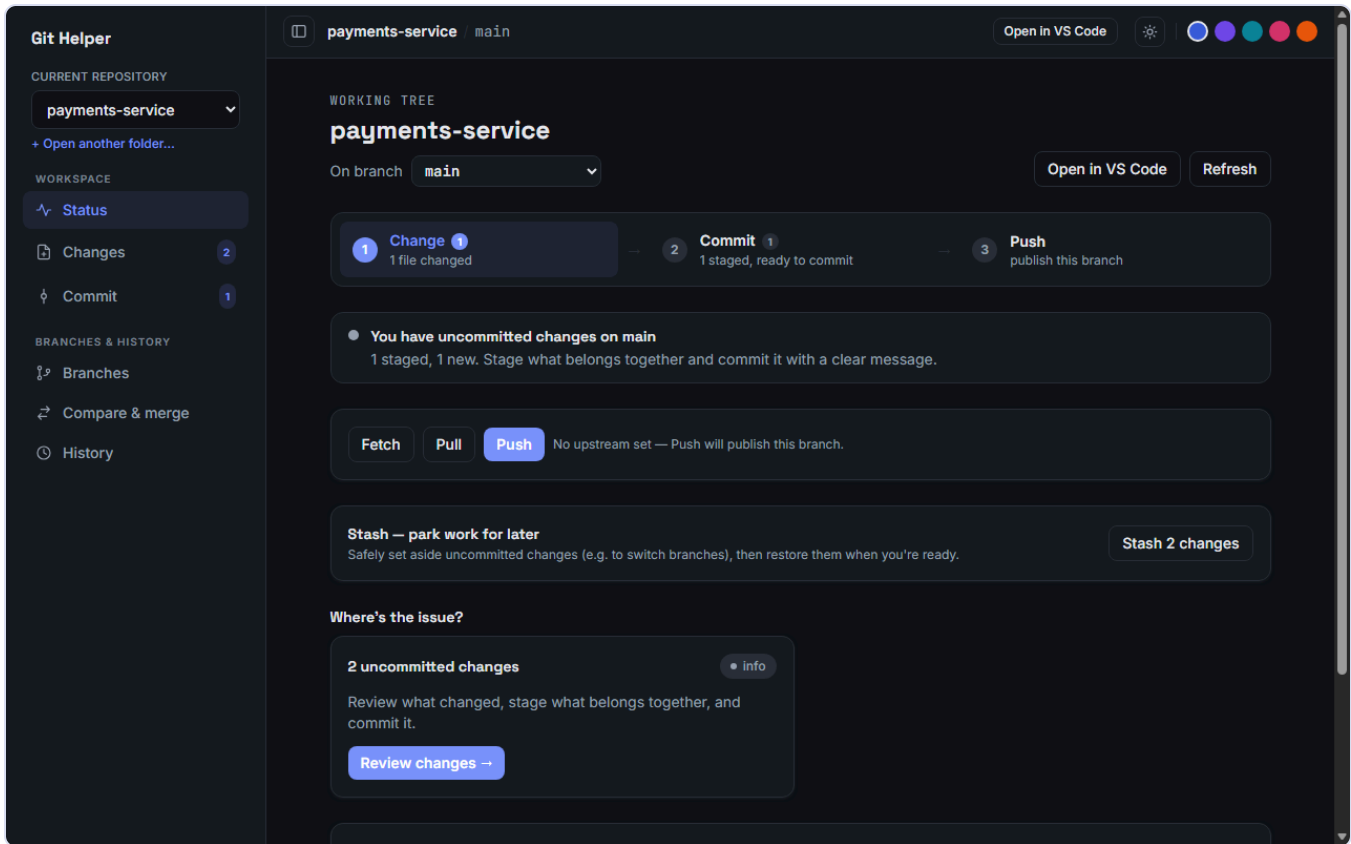
Merging branches *in the cloud* always goes through a pull request — that's the Azure DevOps way. For instant local merges use [Compare & merge](#).

14. Appearance: dark mode & colors

In the top navbar (desktop) or the menu drawer (mobile):

- 🌙 / 🌞 **toggle** — light or dark mode (follows your system setting by default).
- **Five color dots** — pick the app's accent color: Indigo, Violet, Teal, Rose, or Orange.

Both choices are remembered.



The **button** at the far left of the navbar collapses the sidebar to a slim icon rail — hover an icon for its name; click **button** again to expand.

15. AI assistant

The **chat bubble in the bottom-right corner** opens the AI assistant — it can *do* things, not just answer. It works through the same safe operations as the UI: check status, list/create/switch branches, stage and commit with a good message, compare branches, stash, fetch/pull/push, and on the Azure side list branches and PRs and **open a pull request**.

Try: "What's the state of my repo?" · "Create a branch fix/tax and switch to it" · "Stage everything and commit with a sensible message" · "Compare my branch with master and tell me if it's safe to merge" · "Open a PR from my branch into main".

What you can ask it to do

You say (in plain words)	It does
"What changed in my repo?" / "Where do I stand?"	Reads status: branch, ahead/behind, changed files, conflicts
"Commit my work with a good message"	Stages everything and writes a sensible commit message for you
"Make a branch for this fix and switch to it"	Creates + checks out the branch
"Switch back to main"	Checks out the branch
"Is it safe to merge my branch into master?"	Compares the branches and summarizes commits/files/conflict risk
"Set my changes aside" / "bring them back"	Stash and restore
"Get me up to date" / "send my commits up"	Fetch, pull, push (same rules as the buttons)
"List the open PRs" / "open a PR to main"	Azure DevOps: lists branches/PRs, creates a pull request
"Which of my repos have uncommitted work?"	Checks every repository on your list

The same questions work in **Claude Desktop** once connected (below) — and there you also get Claude's general reasoning on top, e.g. *"look at my repos and tell me what I should finish first."*

Safety: the assistant never discards changes, deletes branches, rewrites pushed history, or completes merges/PRs — those stay manual. If it hits a conflict it stops and sends you to the resolver.

One-time setup — pick a provider: open the chat bubble and choose **Claude** or **Gemini**, then paste that provider's API key. It's stored only on your machine and used server-side.

Provider	Where to get a key	Notes
Claude (Anthropic)	<code>console.anthropic.com</code> → API keys	strongest at multi-step git work
Gemini (Google)	<code>aistudio.google.com</code> → Get API key	has a free tier ; both key formats work (<code>AIza...</code> and <code>AQ...</code>)

This works the same on the **hosted web app** — everyone brings their own key, kept in their own encrypted session (private per user, like the Azure token). A site owner can optionally set `ANTHROPIC_API_KEY` / `GEMINI_API_KEY` environment variables to provide a shared key for all visitors instead.

No API key? Connect Claude Desktop (MCP)

If you have the **Claude Desktop app**, you can skip API keys entirely and chat with your repos *inside Claude Desktop* — Git Helper plugs in as an MCP tool provider, and inference runs on your existing Claude subscription.

Anyone who installed Git Helper (including from the downloaded installer):

 **The order matters.** Claude Desktop overwrites this setting if it's running while you connect — quit it first.

- 1. **Quit Claude Desktop completely** (File → Exit / right-click the tray icon → Quit — closing the window is not enough).
- 2. Open **Git Helper** → click the **chat bubble** (bottom-right) → click **Connect Claude Desktop**. (The button warns you if Claude Desktop is still running.)
- 3. **Start Claude Desktop**. In Settings → Developer you should see **git-helper** listed as running.
- 4. In a new chat, ask *"what's the status of my repos?"*, *"commit my changes in payments-service"*, or *"compare my branch with master"*.

If you're connected to Azure DevOps in Git Helper when you click Connect, the Azure tools (list branches/PRs, create a PR) are wired into Claude Desktop too. Claude Desktop sees the same repository list as the app and follows the same safety rules — nothing

destructive, conflicts are reported, not "fixed".

(Running from the repo instead of the installer? `npm run mcp:install` does the same thing from the command line.)

The in-app chat bubble and Claude Desktop work independently — use either or both.

Changing or removing keys, and disconnecting

Click the **gear icon** in the assistant panel's header to open its settings:

- **Switch provider or update the key** — pick Claude or Gemini and paste a new key; it replaces the old one.
- **Remove key** — deletes the stored key (the setup screen returns).
- **Disconnect** (next to Connect Claude Desktop) — removes Git Helper from Claude Desktop. Same rule as connecting: quit Claude Desktop first, or it may restore the entry when it exits.

16. Auto-commit

On the [Status page](#), the **Auto-commit** card lets each repository commit its outstanding changes automatically — off by default, enable it per repo with the toggle. Three schedules:

Schedule	Behavior
Daily	Once a day, commits whatever changed (skips if nothing changed)
Every 2 days	Same, on alternate days
When changes appear (dynamic)	Commits shortly after changes show up (at most every 5 minutes)

Auto-commits are clearly labelled ("*Auto-commit (Git Helper): 3 files — 2026-07-10 14:00*"), and the card shows the last run's result. Safety rules: it **never pushes** (sharing stays your call), and it never touches a repo that is mid-merge, conflicted, or in detached HEAD. Runs while the app (web dev server or desktop app) is open.

17. Undo & restore — take something back

Sidebar → **Undo & restore** (in the WORKSPACE section). One page with every way to take something back, ordered from mild to drastic, each with a plain "When / What it does" description — you never need to know the git command behind it.

#	Undo	When to use it
1	Unstage files	You staged files for the next commit but changed your mind. Edits stay.
2	Stash	Edits aren't ready but you need a clean repo. Nothing is lost.
3	Discard changes	Wrong edits — put files back the way the last commit had them. <i>Permanent</i> .
4	Undo the last commit	You just committed and spotted a problem. The work comes back staged. Only for unpushed commits.
5	Revert a commit	Any commit — even old, even already pushed — was a mistake. Creates a <i>new</i> commit that cancels it, so history is never rewritten. The safe default.
6	Rewind the branch (reset)	The last few <i>unpushed</i> commits shouldn't have happened. Choose whether the undone work stays staged, comes back as edits, or is discarded. Automatically refused if it would rewind pushed commits.
7	Rescue (recent positions)	Something went really wrong. Git keeps every place your branch has been for ~90 days — jump back to any of them, even after a bad rewind.

Shortcuts: in **History**, click any commit and use **↶ Undo this commit** right in the detail panel (or *More undo options* to jump to this page with that commit preselected). If an undo clashes with newer changes, you're taken to the conflict resolver, same as a merge.

On the cloud (Azure DevOps): the Dashboard's *Recent commits* list has an **↶ Undo** button per commit. Azure prepares the opposite change on a new `revert/...` branch and Git Helper opens a pull request with it — nothing on the branch changes until you complete that PR, so an undo is always reviewed and never rewrites shared history.

18. Recovery & sync — the repo check-up

Sidebar → **Recovery & sync**. Think of it as the repo doctor: it **fetches the latest remote info first** (so the numbers are honest), scans for situations that usually confuse people, and shows a card for each one it finds. Every card:

- explains the situation **in plain language**,
- shows the **exact git command** it would run,
- and **never executes anything without a confirm** — truly destructive actions additionally require you to *type a word* (e.g. `discard`) so a reflexive click can't cause damage.

What it detects and offers:

Situation	What the card offers
Local commits not on the remote	Reset the branch to exactly match the remote. If your commits net to zero file changes (add-then-delete noise), it says " <i>Safe — no file changes will be lost</i> "; otherwise it lists in red every file whose local version would be lost
Your last commit is a merge (accidental pull)	Explains that the remote is NOT affected until you push; one click undoes the merge, or keep it and push
Committed too early / wrong message	Points to Undo last commit — changes stay staged, only the commit is undone (blocked once pushed)
Uncommitted changes you don't want	Lists every file, then offers " Stash instead (safe) " first, and "Discard everything" behind a typed confirmation
Commits on a teammate's branch	Detects that most of the branch's pushed history belongs to someone else ("This looks like Dhrumil's branch") and offers to move your commits to a new branch and put theirs back exactly as it was
Detached HEAD ("not on any branch")	Warns that commits made here can be lost; save the spot to a new branch in one click, or jump to an existing branch
Far behind the remote (10+ commits)	Yellow — red at 30+ — "Your copy is outdated by N commits", with plain words on what pulling will do (fast-forward vs merge with your local commits)
Forgotten stashes	"N stashed change sets waiting" with one-click restore — the same badge also appears on the repository picker cards

Safe-push check: when you push a branch whose history mostly belongs to a teammate, the push confirmation adds a warning — "*This looks like ...'s branch*" — and suggests your own branch + a Pull Request instead. You can still push if you two agreed to share the branch.

Action history with undo notes: at the bottom of the page, everything Git Helper did in this repo (pushes, pulls, resets, reverts, recoveries...) is listed newest-first, each entry with a **"How to undo"** note — usually pointing at the Rescue list in **Undo & restore**, which can jump back to any recent position for ~90 days.

19. Troubleshooting

Problem	Fix
"Azure DevOps returned a sign-in page..." when connecting	Check the organization spelling, that the PAT hasn't expired and was created for <i>this</i> organization, and that no company policy (IP restrictions / Conditional Access) blocks the network you're on.
Clone/push fails with authentication error	Your PAT needs Code → Read & write scope. Create a new token and reconnect.
"Open in VS Code" does nothing	Install the <code>code</code> command: VS Code → <code>Ctrl/Cmd+Shift+P</code> → "Shell Command: Install 'code' command in PATH".
Commit fails asking who you are	Set your Git identity once: <code>git config --global user.name "Your Name"</code> and <code>git config --global user.email you@company.com</code> .
Windows: "Windows protected your PC"	Click More info → Run anyway (the installer isn't code-signed).
macOS: "cannot verify the developer"	Right-click the app → Open → Open (one time).
Undo/Amend button is disabled	The last commit is already pushed — rewriting shared history is blocked on purpose. Make a new commit instead.
A branch I see on Azure isn't in my local list	It appears under Branches marked remote — click Check out . If it's brand new, press Fetch first.